Innovative Evaluation Practices for BCSE308P Computer Networks Lab

Fall Semester 2026-27

CCSDS INSPIRED SOFTWARE TELEMETRY RECEIVER AND GROUND STATION

24BCE2055 JAHNAVI SINGH

B.Tech.

in

Computer Science and Engineering

School of Computer Science and Engineering



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github.com/POSTI-25/computer-networks-project

GitHub Repository

CCSDS Inspired Software Telemetry Receiver and Ground Station

Repository URL

https://github.com/POSTI-25/computer-networks-project

The repository contains the software telemetry transmitter, UDP receiver and ground station dashboard. The README provides the software requirements, installation procedure, execution steps and expected demonstration results.

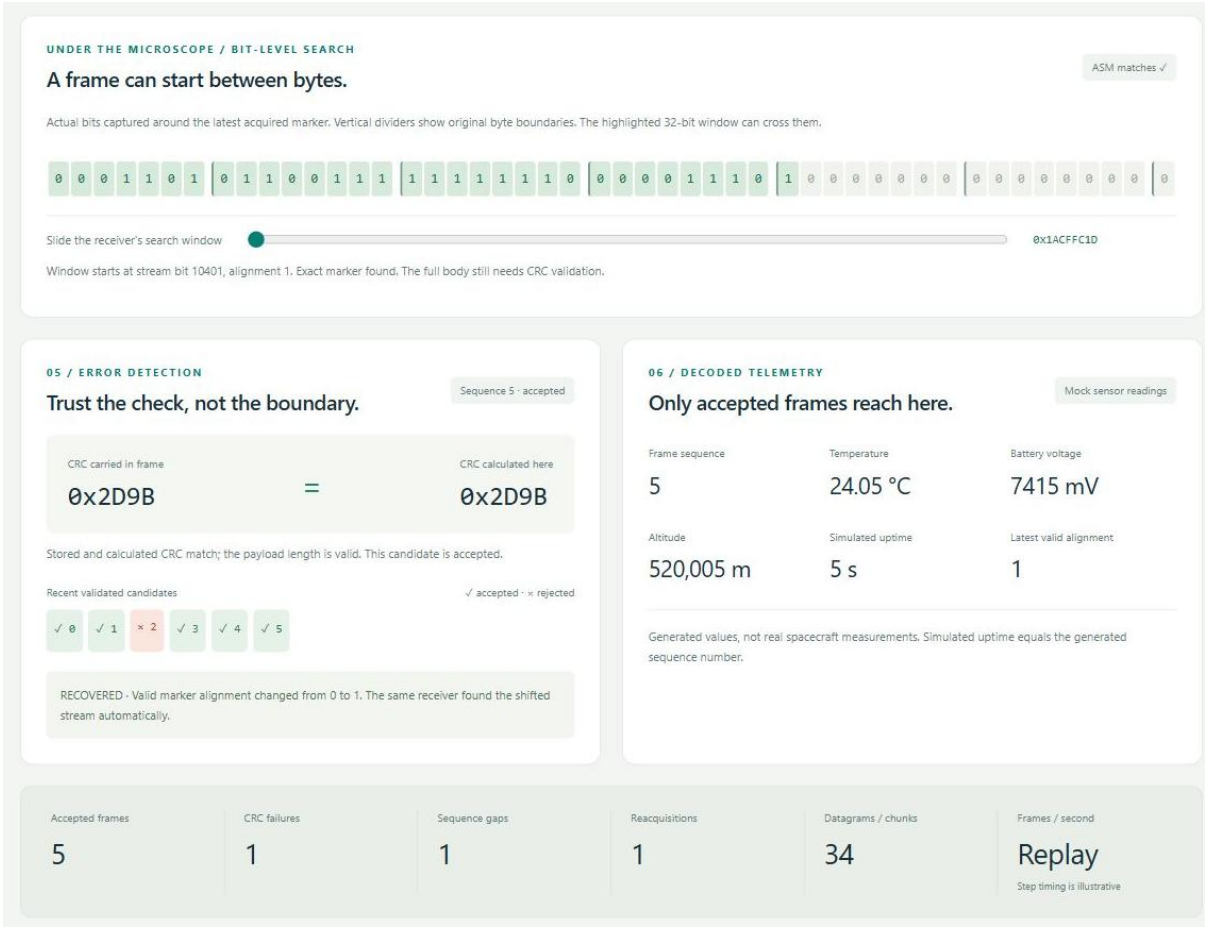
Source and supporting material

Material	Repository location
Implementation	src/ and ui/
Dependencies and configuration	package.json, package-lock.json, .nvmrc and tsconfig.json
Dashboard evidence	docs/screenshots/
Automated tests and saved results	test/ and docs/verification/
Demonstration procedure	README.md and docs/demo-guide.md

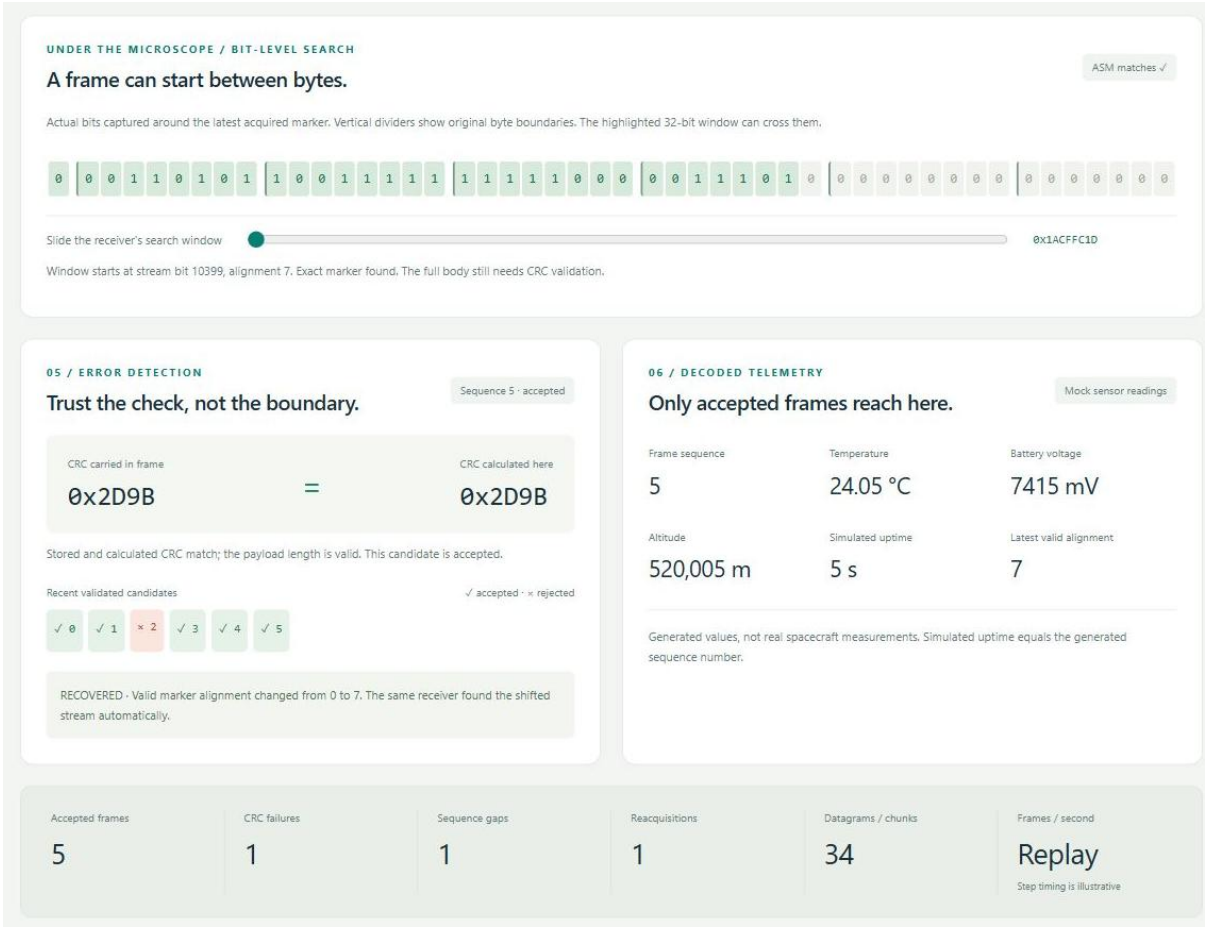
Verification record

25 October 2026: 185 automated tests passed, with no failures or skipped tests. Strict TypeScript checking and the production build also passed using Node.js 22.19.0 and npm 10.9.3.

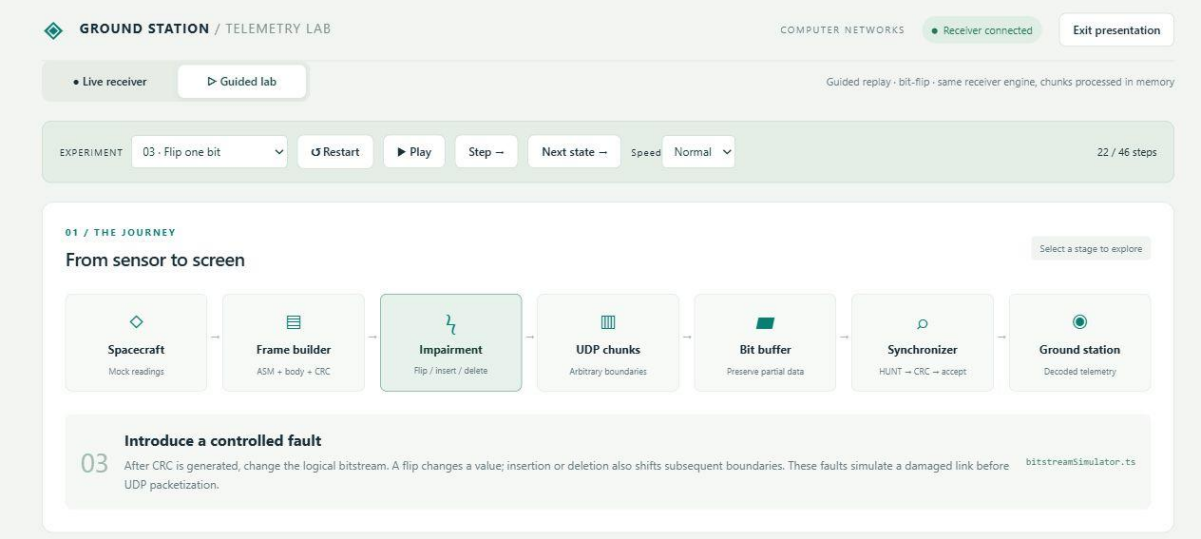
This project uses a software bitstream simulator and real localhost UDP communication. Cisco Packet Tracer files and database files are not applicable to this implementation.



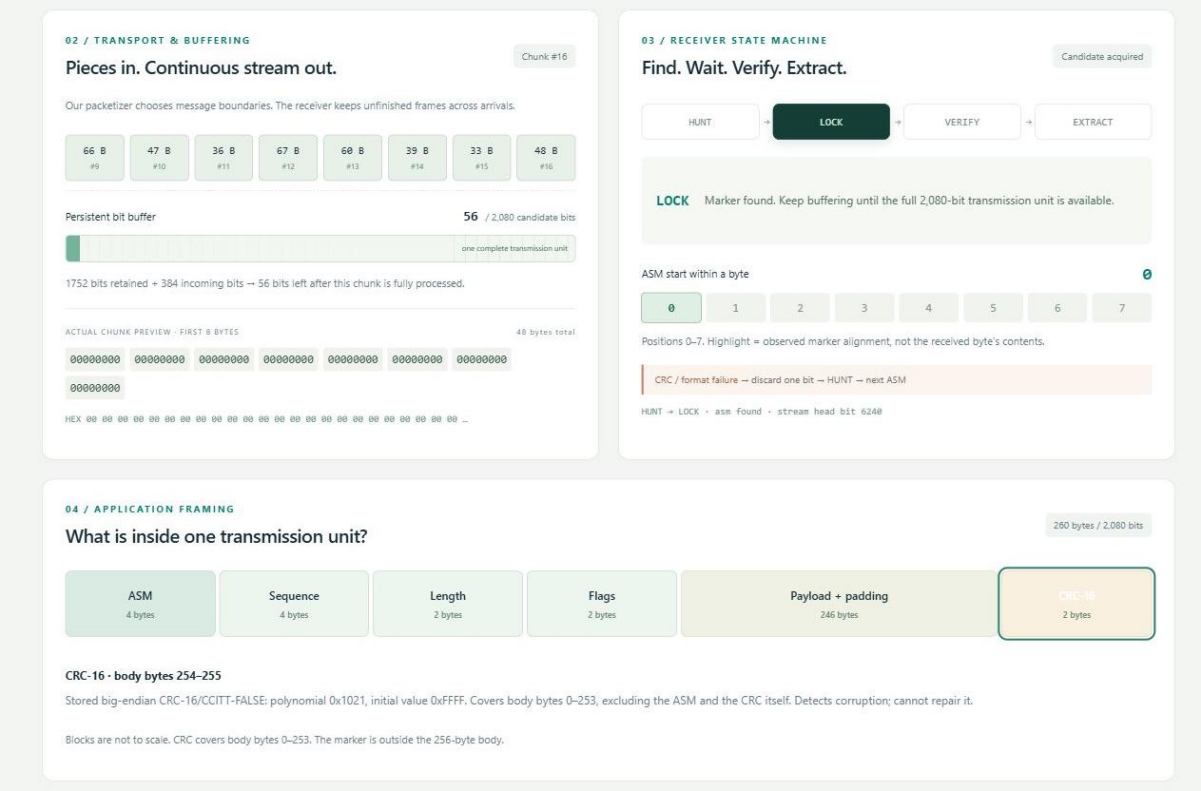
Guided bit insertion demonstration showing frame reacquisition at alignment 1.



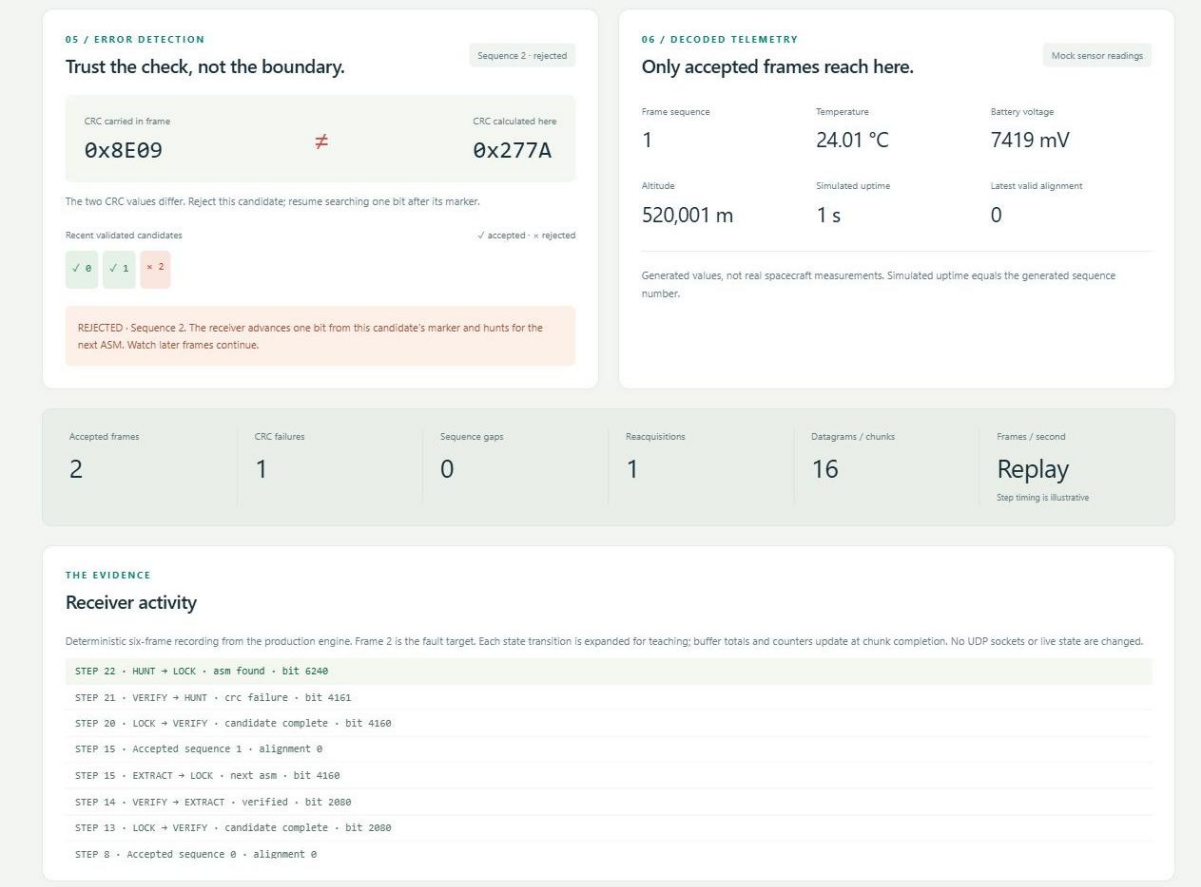
Guided bit deletion demonstration showing frame reacquisition at alignment 7.



Transmission pipeline and guided experiment controls.



Persistent buffer, receiver state and frame structure.



CRC mismatch and rejection of sequence 2.

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